

COP3330C Module 9 Programming Assignment 1: Applying the Adapter Pattern with Java Collections

In this assignment, you will solidify your understanding of Java collections, generics, and design patterns by creating a Java program that demonstrates the use of the Adapter pattern, along with other Java concepts such as the Diamond Operator, sorting a list of numbers, using the `remove()` method, and employing unbounded wildcards. Note that this assignment requires the submission of a source code file and a screenshot of command line execution (not from an IDE).

Instructions:

- **Setup Your Project:**
 - Create a new Java class named `ConceptDemo`.
 - Write a `main()` method to implement all of the required tasks.
- **Apply the Adapter Pattern:**
 - Create an `ArrayAdapter` class that wraps an `Integer[]` and provides a `toList()` method that returns a new, mutable `List<Integer>` (implemented as an `ArrayList<Integer>`).
 - Create an `Integer[]` array with several elements (e.g., 5, 2, 9, 1, 4, 7).
 - **Note:** We are using an Object Adapter (rather than a Class Adapter). The `ArrayAdapter` contains an `Integer[]` as a component and adapts it to the `List<Integer>` interface instead of exposing the array directly.
- **Use the Diamond Operator:**
 - In the `toList()` method, create the `ArrayList<Integer>` using the Diamond Operator (`<>`).
 - Use `Collections.addAll()` to populate the list with the elements from the array.
- **Sort the List:**
 - In the `main()` method, call the adapter's `toList()` method and store the returned `List<Integer>`.
 - Print the list before sorting.
 - Use `Collections.sort()` to sort the elements of the list.
 - Print the sorted list to verify the results.
- **Use the `remove()` Method:**
 - Remove a specific element from the list by value (e.g., remove the value 4). Use `Integer.valueOf()` to avoid removing by index.
 - Print the list again to show the result after the removal.
- **Demonstrate Unbounded Wildcards:**
 - Create a method named `printList()` that accepts a `List<?>` parameter.
 - Call `printList()` from the `main()` method, passing the `List<Integer>` created by the adapter.
 - Use the `printList()` method to display the list's contents..
- **Testing:**
 - Your program should display labeled output showing:

- The list before sorting.
- The list after sorting.
- The list after removing the value 4.
- The list printed by calling the printList() method.

Program Output Example:

```
Before sorting: [5, 2, 9, 1, 4, 7]
After sorting: [1, 2, 4, 5, 7, 9]
After removing 4: [1, 2, 5, 7, 9]
Printing list using unbounded wildcard: [1, 2, 5, 7, 9]
```

Functional Requirements:

- Adapter Pattern: Correctly adapt an Integer[] array to a List<Integer>.
- Diamond Operator must be used for list creation.
- Sorting must be demonstrated using the Collections.sort() method.
- Remove an element from the list by specifying the value.
- Unbounded wildcard must be used in the printList() method.

Non-Functional Requirements:

- 4-line ID Header (top of the source file(s)), <= 80 columns each:
 - // FileName.java
 - // your name
 - // submission date
 - // one-line summary of program's purpose
- All lines <= 80 columns (including comments)
- Use of appropriate comments in your code to explain each major step.
- Correct indentation (minimum of 3 spaces per level)

Submission:

- Submit your ConceptDemo.java file to the GitHub repository.
- Submit a screenshot (.png or .jpg only) of the program output demonstrating the steps as shown above. The screenshot must be taken of execution **from the command line**, not from your IDE. It should be clear and legible.
- Do not submit .class files.